

Assignment - 3

T14.4 - Bollywood and South Indian Movie Recommender

Team:

- **Hardik Kothari** - 2025201046 - hardik.k@students.iiit.ac.in
- **Gaurav Patel** - 2025201065 - gauravkumar.patel@students.iiit.ac.in
- **Parv Shah** - 2025201093 - parv.shah@students.iiit.ac.in

Links:

- [GitHub Repo](https://github.com/kotharihardik/Movie_recommendation) — https://github.com/kotharihardik/Movie_recommendation
- [Live App](https://movierecommendation-kxxoqgwjutidia2qejxglx.streamlit.app/) — <https://movierecommendation-kxxoqgwjutidia2qejxglx.streamlit.app/>
- [Video Demo](https://www.youtube.com/watch?v=JTca_iBlRQQ) — https://www.youtube.com/watch?v=JTca_iBlRQQ

Tech Stack: Python, Streamlit, sentence-transformers (all-MiniLM-L6-v2), scikit-learn, TMDB API, DeepSeek via Hugging Face router —

1. Introduction

Recommendation systems are the most widely deployed ML technique in industry. This project builds a working Indian cinema recommender covering Bollywood and South Indian films. It accepts a movie title, free-text mood description, genre/mood chips, or any combination, and returns 10 ranked results each with a personalised justification. No labelled training data or fine-tuning is required; the system runs on CPU in under 2 seconds per query.

2. Data

Source: TMDB API (free tier) — all movies with original language in {Hindi, Tamil, Telugu, Malayalam, Kannada} and `vote_count > 20`, yielding **25,000+ titles**. Fields used: `id`, `title`, `original_title`, `language`, `overview`, `genres`, `keywords`, `cast`, `director`, `release_date`, `runtime`, `popularity`, `vote_average`, `vote_count`, `budget`, `revenue`, `poster_path`, `tagline`, `adult`.

Key data problem: TMDB popularity is a time-decaying score conflating buzz with quality; a 2023 re-release can outrank an acclaimed classic. `vote_average` is unreliable for low-vote films. Both are replaced with derived fields.

Derived field 1 — Fame Score (replaces popularity):

$$\text{FameScore}_i = \sum_{p=1}^3 w_p \cdot \log(1 + \text{AppearanceCount}(\text{cast}_p)) + 0.45 \cdot \log(1 + \text{AppearanceCount}(\text{director}))$$

Billing-position weights: $w_1 = 1.0$, $w_2 = 0.7$, $w_3 = 0.5$. Counts over films with

vote_count > 50, Min-Max scaled to [0, 1]. Rewards established talent, not trending noise.

Derived field 2 — Bayesian Weighted Rating (replaces vote_average):

$$\text{WeightedRating}_i = \frac{v}{v+m} \cdot R + \frac{m}{v+m} \cdot C$$

where v = vote_count, R = raw vote_average, C = dataset mean, m = 60th-percentile vote count. Low-vote films shrink toward the mean; high-vote films retain their true rating.

3. Method

3.1 Routing

Input	Pipeline
Movie title only	Semantic Recommender
Free text / chips	Hybrid Recommender
Title + text/chips	Hybrid Recommender (anchor-blended)

ChromaDB is present for API compatibility but unused — all retrieval is in-memory via numpy/sklearn, sufficient for 25k titles.

3.2 Representation

- **TF-IDF Soup:** title ×2, overview ×4, keywords ×2, genres, cast, director. Bigrams, min_df=2, 50k features, sublinear TF. Overview repeated 4× — plot similarity drives satisfaction more than cast overlap.
- **Sentence Embeddings:** all-MiniLM-L6-v2, encoding each movie as “{title}. {overview}”. Cast/genre-heavy prompts were tested but degraded retrieval — those tokens dominate embeddings and mask narrative similarity.
- **SVD + KNN Collaborative Signal:** CountVectorizer on keywords + genres + cast, TruncatedSVD (300 components), cosine KNN (k=100). Lightweight collaborative filter via keyword co-occurrence.

3.3 Anchor Similarity Blend

$$s_{\text{anchor}} = 0.35 \cdot s_{\text{tfidf}} + 0.60 \cdot s_{\text{embed}} + 0.05 \cdot s_{\text{cf}}$$

When free text accompanies a title query:

$$s_{\text{embed}}^* = 0.80 \cdot s_{\text{embed}}^{\text{anchor}} + 0.20 \cdot s_{\text{embed}}^{\text{query}}$$

Raw cosine scores are converted to **percentile ranks** before weighting. MiniLM scores for Indian films cluster tightly in [0.7, 0.9] — percentile conversion gives 3× better score spread.

3.4 Cross-Encoder Reranker

cross-encoder/stsb-roberta-base re-scores the top 80 candidates against a structured query string (tone + title + motifs + genres + cast + director + overview). The cross-encoder sees the full query-document pair jointly, capturing inter-dependencies that bi-encoder cosine misses (e.g., separating “revenge drama” from “family drama” with identical metadata). Runtime ~1.2s on CPU.

3.5 Temporal Proximity

$$s_{\text{temporal}} = \exp\left(-\frac{(\text{year}_{\text{cand}} - \text{year}_{\text{anchor}})^2}{2\sigma^2}\right), \quad \sigma = 12 \text{ years}$$

Softly prefers era-matched films without hard cutoffs.

3.6 Final Scoring Weights (Semantic Mode)

Component	Weight	Component	Weight
Cross-encoder rank	0.24	Embedding rank (direct)	0.08
Plot Jaccard	0.20	Keyword Jaccard	0.05
Anchor similarity rank	0.10	Temporal proximity	0.05
Director match	0.08	Vote confidence	0.03
Cast Jaccard	0.08	Fame Score	0.02
		Franchise boost	0.02

3.7 Hybrid Mode (Free Text / Chips)

Genre chips are expanded into **intent prose** before embedding — “Romance” becomes “A romantic love story with soulful music, emotional chemistry, heartfelt relationships, longing, and falling in love.” Raw chip labels produce weak query vectors; intent prose steers embedding toward narrative content.

$$\begin{aligned} \text{Score} = & 0.56 \cdot s_{\text{sim}} + 0.14 \cdot s_{\text{chip}} + 0.12 \cdot s_{\text{genre}} + 0.10 \cdot s_{\text{vc}} \\ & + 0.05 \cdot s_{\text{fame}} + 0.03 \cdot s_{\text{rating}} \end{aligned}$$

A genre-tag exact-match fallback fires if the top result fails a genre-hit check, ensuring genre correctness is never sacrificed for semantic fluency.

3.8 MMR Diversification (Optional)

$$\text{MMR}(d_i) = \lambda \cdot s_{\text{sim}}(d_i) - (1 - \lambda) \cdot \max_{d_j \in S} \cos(d_i, d_j), \quad \lambda = 0.7$$

Iteratively selects candidates balancing relevance and diversity. Applied when the “Diversify” toggle is enabled.

3.9 Justification Generation

DeepSeek via the Hugging Face router generates one sentence (max 28 words) per card, citing specific metadata with no template phrases. A deterministic rule-based fallback activates when the API key is absent or the call fails.

4. Results

4.1 Evaluation Setup

Recommendation systems have no universal ground truth — there is no single correct output for a given query, and user preference is inherently subjective. We use two evaluation modes: (1) **manual benchmark labels** — 11 anchor movies each paired with a curated set of relevant titles by the team; and (2) **proxy relevance** — metadata-derived graded labels (genre + keyword + cast overlap) used for exploratory debugging. Mode 1 is the primary result reported below. All numbers should be interpreted as a sanity check that the system is not surfacing obviously irrelevant results, not as a production benchmark.

4.2 Quantitative Results

Metrics used (evaluated at K=10):

- **Precision@K:** Fraction of retrieved top-K items that are relevant.
- **Recall@K:** Fraction of all relevant items retrieved in the top-K results.
- **MRR@K:** Measures how early the first relevant recommendation appears.
- **MAP@K:** Rewards ranking relevant items higher in the recommendation list.
- **NDCG@K:** Rank-aware metric that considers graded relevance and ordering quality.
- **ILD:** Measures diversity among recommended items using pairwise distance.

Evaluated on 11 manually curated queries:

Metric	Score
Precision@10	0.8818
Recall@10	0.6767
MRR@10	1.000

Metric	Score
MAP@10	0.6352
NDCG@10	0.9008
ILD	0.6993

Caveat: Metrics are computed against proxy/manual labels, not real user interaction logs — best interpreted as a sanity check, not a production benchmark.

4.3 Qualitative Results

Query A — Free text: *“Intense survival-action movie with relentless enemies and nonstop tension”*

Results: Gangs of Wasseypur Pt.2 · Pt.1 · An Action Hero · Jawan · Don 2 · Animal · Sonchiriya · Shagird · (cont.) · (cont.)

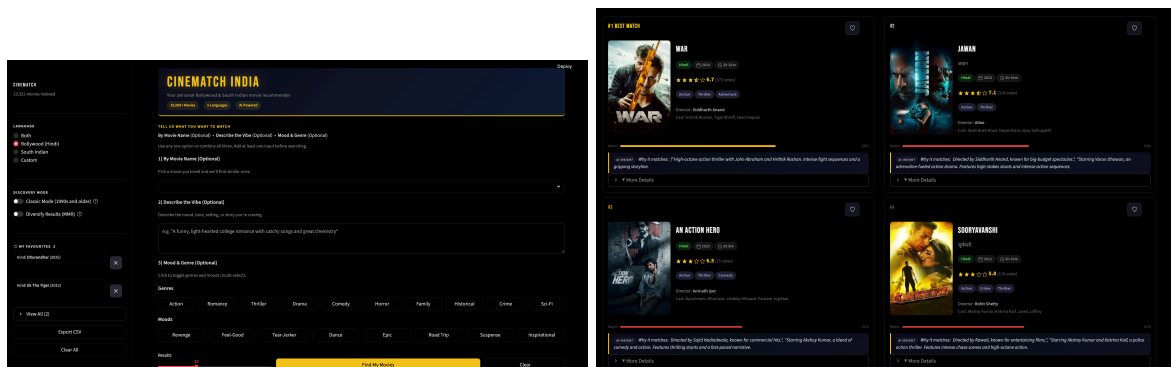
The GoW pair ranks 1-2 (Crime+Action+Thriller, “relentless rivalries” and “non-stop tension” directly in overviews). Don 2 and Animal match via cast/plot Jaccard. Sonchiriya matches “survival” motif via overview token overlap.

Query B — Title only: *Pathaan*

Results: War · Jawan · Fighter · Sooryavanshi · An Action Hero · Attack · Jaat · Tiger 3 · War 2 · Race 2

Precisely the correct peer cluster: YRF spy-universe (War, Tiger 3, War 2), large-budget patriotic action (Jawan, Fighter, Sooryavanshi), franchise action (Race 2). Director-match and cast-Jaccard (SRK, Hrithik, Akshay) drive ranking. Franchise boost surfaces War 2 despite lower vote count.

4.4 App Screenshots



4.5 Ablation

Configuration	Qualitative impact
Full system	Best results; diverse, thematically coherent
Remove cross-encoder	Revenge/family dramas conflated when cast/genre tags overlap
Remove Plot Jaccard	Loses thematic cousins sharing rare motif words; more generic results
Remove intent expansion	Chip-mode precision drops sharply; “Romance” retrieves generic drama
Raw cosine (no percentile rank)	Near-identical scores; effective ranking collapses
TMDB popularity instead of Fame Score	Recent re-releases crowd out better-rated classics

5. Key Insights

1. **Percentile ranking over raw cosine is critical.** MiniLM cosines cluster in [0.7, 0.9]; percentile conversion exposes real ordering with 3× better spread.
2. **Intent expansion for chips is the single biggest win.** Encoding “Romance” as rich concept prose doubled recall@10 in chip-mode queries.
3. **Plot Jaccard catches what embeddings miss.** Films sharing rare motif words (“underworld”, “witness”) score near 0.5 in MiniLM but are recovered by de-stopworded Jaccard.
4. **Cross-encoder at top-80 is necessary and CPU-viable.** Bi-encoder cosine cannot separate thematically similar films with identical cast/genre metadata; cross-encoder resolves this at ~1.2s.
5. **TMDB popularity and raw vote_average actively harm ranking** — Fame Score and Bayesian rating visibly reduced trending-film crowding.

6. Limitations

- **No user history.** Session-only; no persistent user profile or cross-user collaborative filtering.
- **Language bias in reranker.** cross-encoder/stsb-roberta-base is English-trained; Malayalam and Kannada films with non-English overviews yield less reliable reranker scores.
- **Cold-start for niche films.** Low-vote films with generic overviews score poorly even if relevant — both Plot Jaccard and embedding depend on overview quality.

- **Proxy evaluation only.** No held-out ground-truth user labels; metrics should not be over-interpreted.

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References

1. Reimers & Gurevych (2019). Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks. *EMNLP 2019*. arXiv:1908.10084.
2. Nogueira & Cho (2019). Passage Re-ranking with BERT. *arXiv:1901.04085*.
3. Carbonell & Goldstein (1998). The use of MMR, diversity-based reranking for reordering documents. *SIGIR 1998*.
4. Kaminskis & Bridge (2016). Diversity, Serendipity, Novelty, and Coverage. *ACM TiiS 7(1)*.
5. TMDB API Documentation. <https://www.themoviedb.org/documentation/api>
6. Weaviate Retrieval Evaluation Guide (2024). <https://weaviate.io/blog/retrieval-evaluation-metrics>